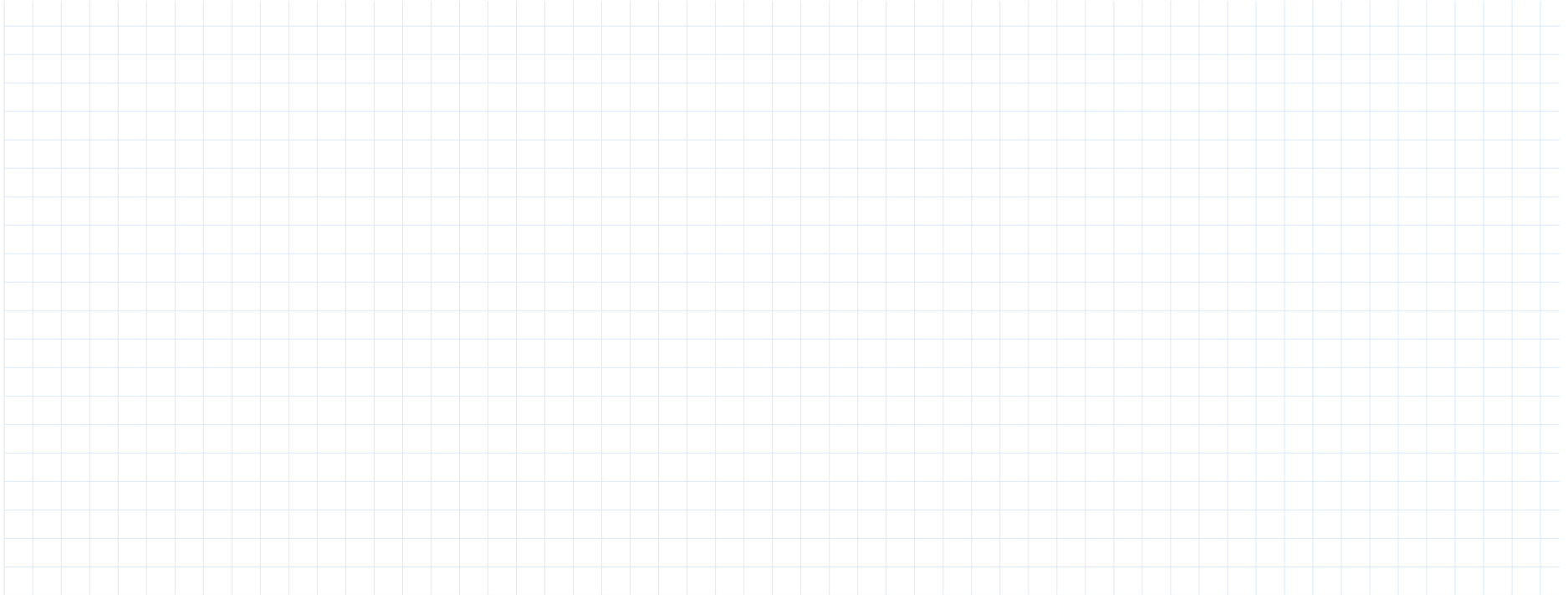


Products and Quotients Inside a Logarithm

For $u, v > 0$ and $a > 0, a \neq 1$:

$$\log_a(uv) = \log_a u + \log_a v, \quad \log_a\left(\frac{u}{v}\right) = \log_a u - \log_a v.$$

Evaluate exactly: $\log_2 4 + \log_2 8$ and $\log_5 75 - \log_5 3$.



Powers and Inverse Properties

For $u > 0$, $\log_a(u^r) = r \log_a u$. Also $\log_a 1 = 0$, $\log_a a = 1$, and $a^{\log_a u} = u$. Simplify exactly:

(a) $\ln(e^5)$ (b) $4^{\log_4 3 + \log_4 2}$ (c) $\log_7 1$.

Use $a^r = e^{r \ln a}$ to rewrite 3^{2x} using base e .



Expand and Condense

Assume $x, y, z > 0$. Expand the first expression as far as possible; condense the second into one logarithm.

$$(a) \ln\left(\frac{x^3\sqrt{y}}{z}\right) \quad (b) 2\log_3 x - \frac{1}{2}\log_3 y.$$

Name the property used at each step.

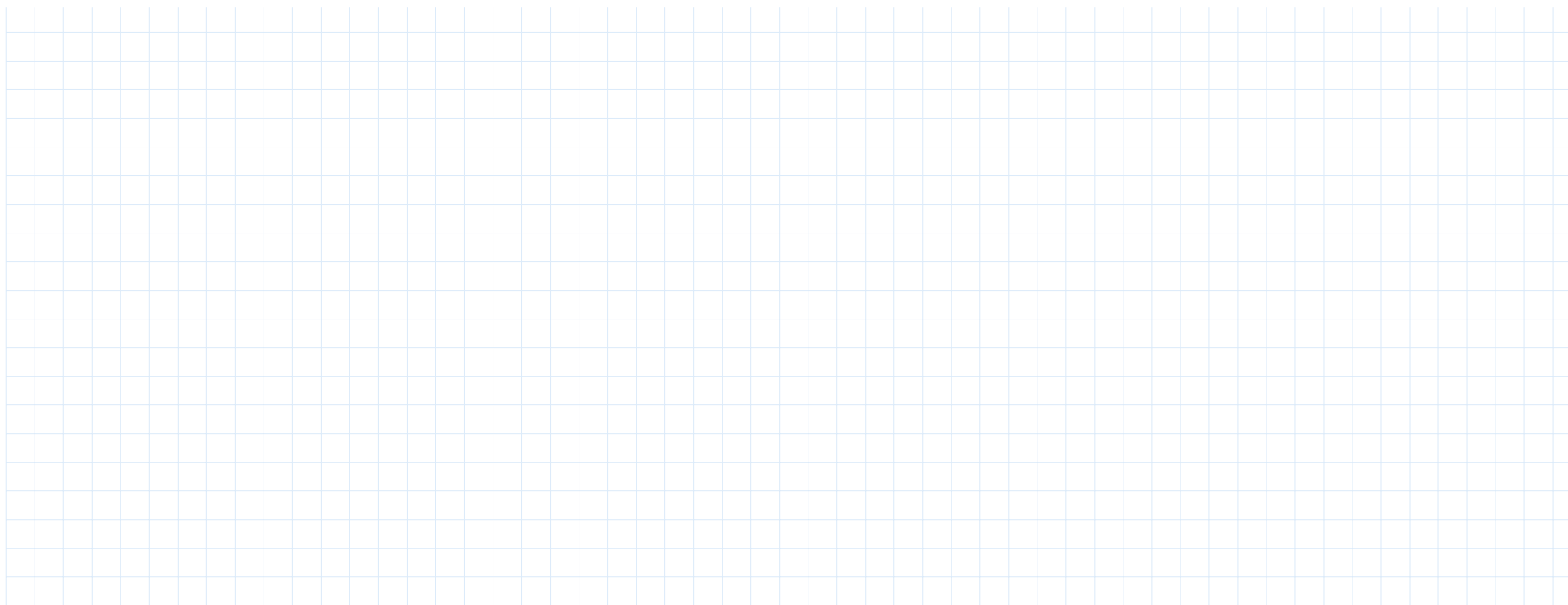


Change of Base

For $M > 0$ and valid bases a, b ,

$$\log_a M = \frac{\log_b M}{\log_b a}.$$

Write $\log_3 7$ using natural logarithms and approximate to three decimal places. Then simplify $(\log_2 3)(\log_3 8)$ exactly.



Solve, Then Check the Original Domain

Solve over the real numbers:

$$\log_3 x + \log_3(x - 24) = 4.$$

Write the domain restrictions first. Combine the logarithms, solve the resulting equation, and check every candidate.

